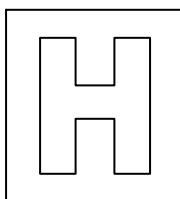


Candidate Name: _____

Class Adm No

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2016 Preliminary Examination II

Pre-University 3

H2 Biology

9648/03

Applications Paper and Planning Question

22 September 2016

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Admission number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.
You will lose marks if you do not show your working or if you do not use appropriate units.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
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Total	

This question paper consists of 21 printed pages including 1 blank page

[Turn over

Answer **all** questions.

1. Before the emergence of more advanced technologies, cDNA libraries have been used to study the genetic changes involved in cancer development.

Normal and tumour cells are obtained from the same patient. Reverse transcription is carried out on mRNA isolated from the tumour cells. Primers consisting of thymine repeats were used during reverse transcription to form single-stranded cDNA as depicted in Figure 1.1. These cDNA are then labelled with fluorescent dyes.

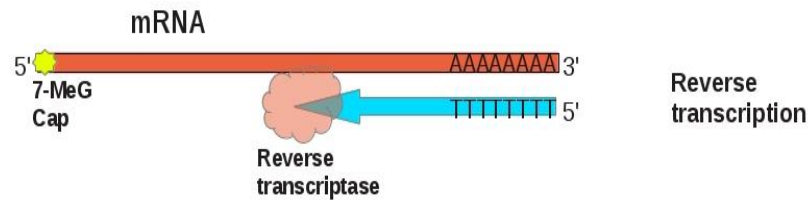


Figure 1.1

Normal cell mRNA and tumour cell cDNA are allowed to hybridise; the resulting double-stranded hybrid molecules and remaining single-stranded mRNA are discarded. Subtracted cDNA (also known as non-hybridized cDNA) are used to form a subtracted cDNA library. The process is summarized by Figure 1.2.

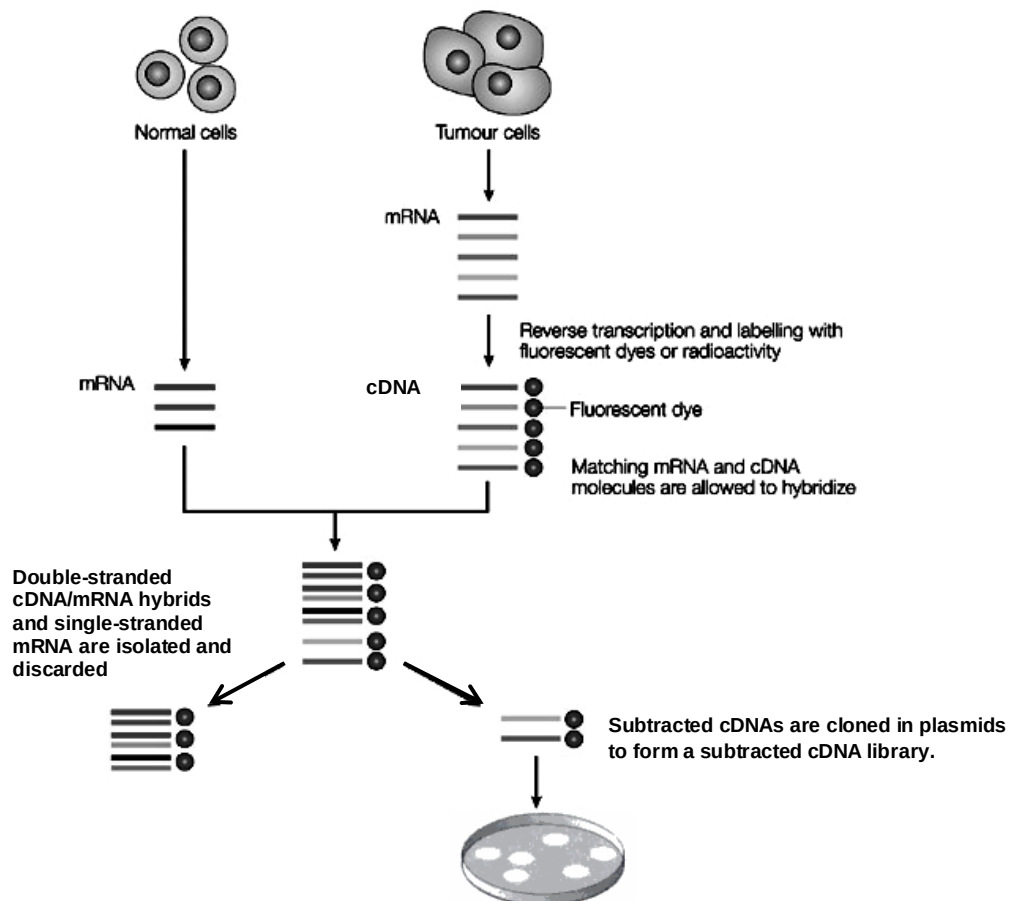


Figure 1.2

A young scientist conducting this procedure was concerned that possible presence of bacteria within cell samples might result in production of bacterial cDNA. This would contaminate downstream processes. However, her concerns were dismissed by her partners who assured her that reverse transcription of bacterial mRNA is unlikely to occur.

- (a) Using the information given, explain why reverse transcription of bacterial mRNA is unlikely to occur.

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- (b) Describe the steps needed to create recombinant DNA molecules that are used to assemble a plasmid library of subtracted cDNA.

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- (c) Explain why the DNA sequences in the subtracted cDNA library are considered mutant alleles implicated in cancer.

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- (d) Suggest why the subtracted cDNA library may not fully capture all possible DNA sequences implicated in cancer.

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Screening of the subtracted cDNA library of a cancer patient revealed a mutated protein kinase gene. A research team decided to clone this gene to isolate the mutant protein and study it to better understand its role in cancer development.

The protein kinase gene was first isolated. The artificial plasmid, pKY350, was constructed to act as an expression vector.

The plasmid was constructed to include two genes, each giving resistance to a different antibiotic: an ampicillin resistance gene and a tetracycline resistance gene. The plasmid also has a target site for the restriction enzyme, *Bam*HI, in the middle of the tetracycline resistance gene.

A pKY350 plasmid was cut using *Bam*HI and the cDNA sequence for the mutant protein kinase was inserted into it. Figure 1.3 shows pKY350 and the recombinant plasmid.

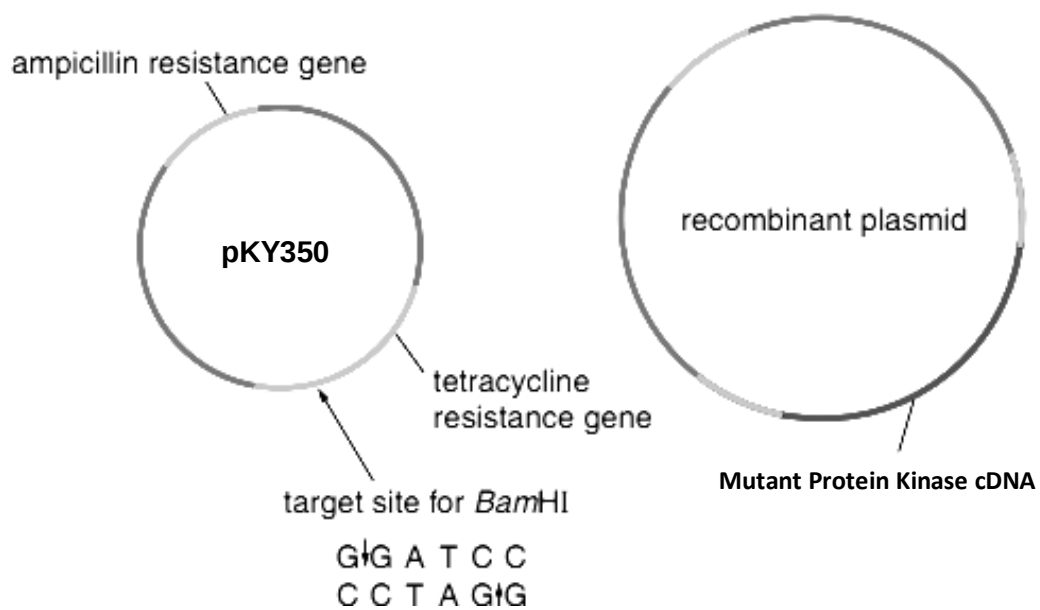


Figure 1.3

The mixture containing recombinant DNA was used to transform *E.coli* bacteria. Replica plating was used to identify recombinant bacteria with the mutant protein kinase gene. Figure 1.4 shows the bacterial colonies that grew on different nutrient agar plates.

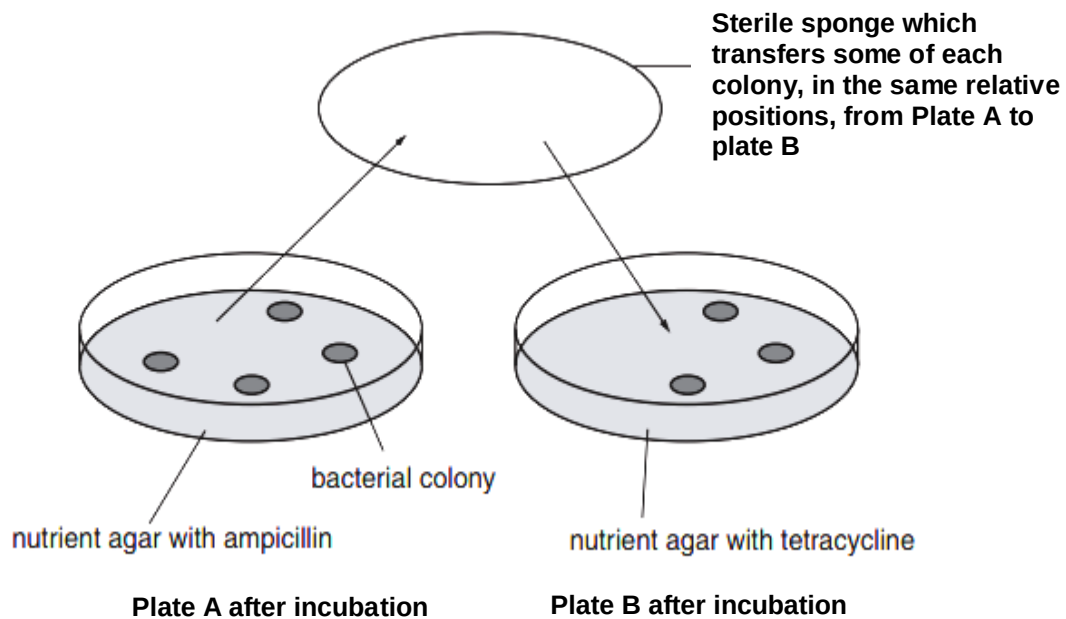


Figure 1.4

- (e) Use a label line and the letter C to identify, on Figure 1.4, a colony of bacteria that contain the recombinant plasmid.

[1]

A senior scientist declared that it would be a challenge to develop a good understanding of the role of the mutant protein kinase in cancer development from studying the recombinant protein produced by a bacterial cloning system.

- (f) Explain why recombinant mutant protein kinase produced by a bacterial cloning system may not be a good model of study for understanding cancer development.

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Today, due to the findings of the human genome project, polymerase chain reaction (PCR) has largely replaced cDNA libraries to directly isolate genes in the study of diseases. PCR is normally carried out using the enzyme Taq DNA polymerase. This enzyme was originally extracted from the bacterium *Thermus aquaticus*. This bacterium was found in Mushroom Spring, one of the hot springs in Yellowstone National Park in the USA.

Taq DNA polymerase is now obtained from genetically modified *Escherichia coli* that carry the Taq DNA polymerase gene. The optimum temperature of this enzyme is 75-80°C.

After 40 minutes at 95°C, the activity of the enzyme is reduced by half. This means that the half-life of this enzyme at 95°C is 40 minutes.

- (g)** Using your knowledge of PCR, explain why a half-life of 40 minutes at 95°C allows many cycles of PCR before the enzyme needs to be replaced.

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 [3]

- (h)** Suggest why Taq DNA polymerase is now obtained from genetically modified *E.coli*.

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 [2]

Work on the human genome project was able to provide many benefits to the field of molecular medicine.

- (i)** State 2 other areas that have benefited from work on the human genome project.

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 [2]

[Total: 19]

2. Pluripotent stem cells are one type of stem cells and there is widespread interest in their study today. One key reason is because pluripotent stem cells can be induced to differentiate into the specific cell type required to repair damaged or destroyed cells or tissues. Additionally pluripotent stem cells can be used to study early events in human development and find out more about how cells differentiate and function. This may help researchers find out why some cells become cancerous and how some genetic diseases develop.

In 2006, a Japanese Scientist, Shinya Yamanaka, made a ground-breaking finding that would win him the Nobel Prize in Physiology or Medicine just six years later. He discovered that specialized cells can be stimulated to dedifferentiate and change back into pluripotent stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In experiments with mice, Yamanaka showed that the introduction of four genes caused specialized cells to change to iPS cells. Further studies have shown that these four genes indirectly result in the reprogramming of specialized cells by influencing the activity of many other genes.

- (a) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of specialized cells into iPS cells.

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Pluripotent stem cells can also be derived from cytoplasmic hybrid (cybrid) cells through the method of somatic cell nuclear transfer (SCNT). A procedure for producing pluripotent stem cells by SCNT is outlined in Figure 2.1.

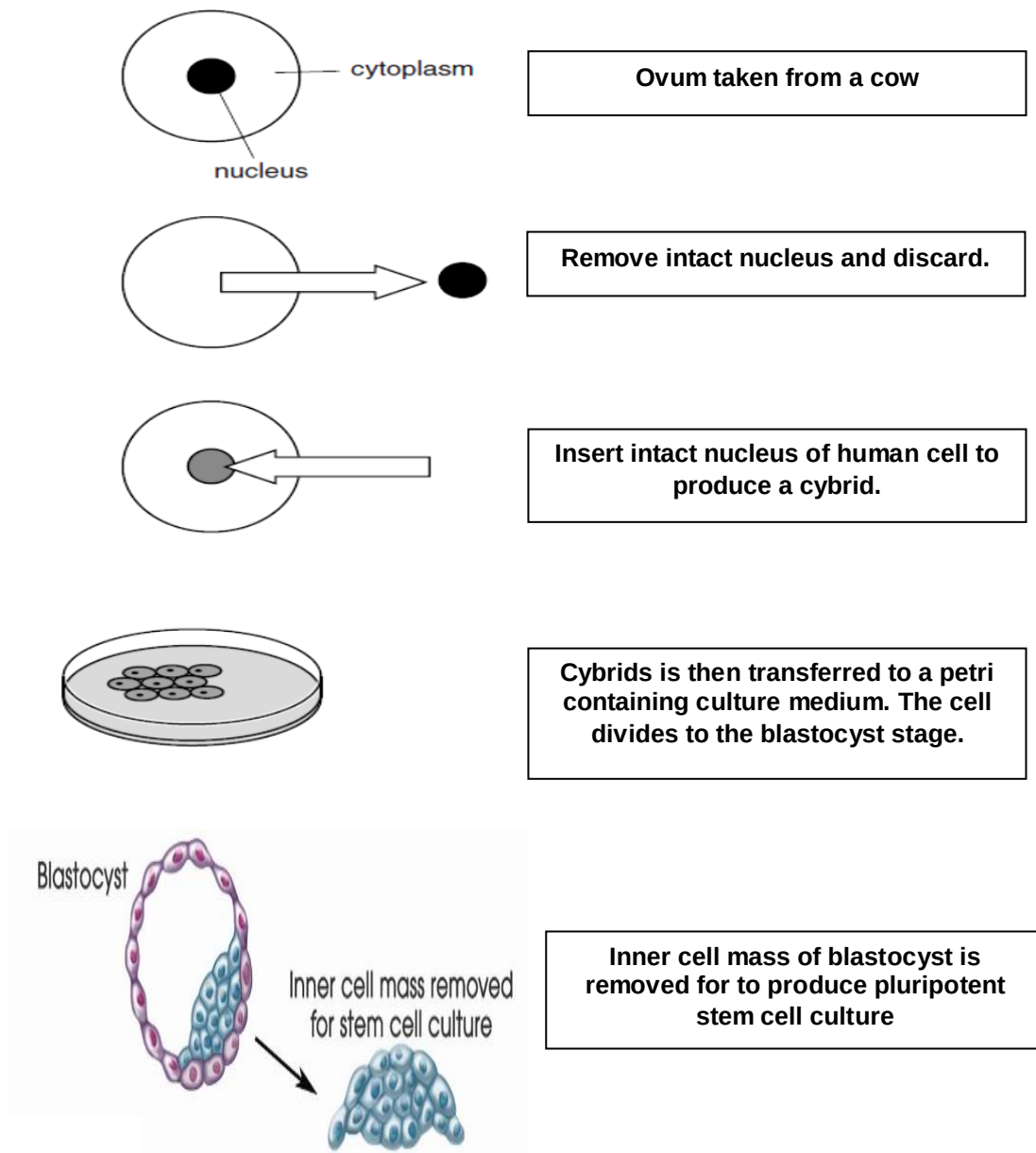


Figure 2.1

The DNA from the pluripotent stem cells generated by the above method is found to be 99.6% human.

(b) Explain the origins of the 0.04% cow DNA that is found in these cells.

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Some people have argued that it is unethical to allow the production of cybrids.

- (c) Suggest why the production of hybrids may be considered unethical.

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 [1]

In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. Adult stem cells derived from the woman's own bone marrow were used. Samples of these stem cells, together with cells derived from the lining of her trachea, were placed in a bioreactor for four days during which time they multiplied to coat a collagen framework. This cell coated framework was then used to replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

- (d) State two reasons why using the patient's own stem cells to treat a damaged bronchus is preferred to a transplant involving another donor.

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 [2]

- (e) Describe the unique features of the adult stem cells derived from the bone marrow.

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 [3]

[Total: 10]

3. Iron-deficient anaemia is one of the most serious problems worldwide. Some scientists decided to explore the use of transgenic rice with increased copies of the Nicotianamine Synthase (NAS) gene to address the problem of iron deficiency. NAS is an enzyme in the metabolic pathway involved with iron acquisition in the rice plant. In these regards, the use of transgenic rice to address human iron deficiency is based on developing rice with increased iron content as transgenic rice would contain more copies of the NAS gene and produce more NAS enzymes compared to wild type rice.

To produce transgenic rice, a recombinant Ti plasmid containing NAS gene is engineered and reintroduced back into the bacteria *Agrobacterium tumefaciens*. The recombinant bacteria is then allowed to infect the plant. The process showing how transgenic rice plant is produced is illustrated in Figure 3.1.

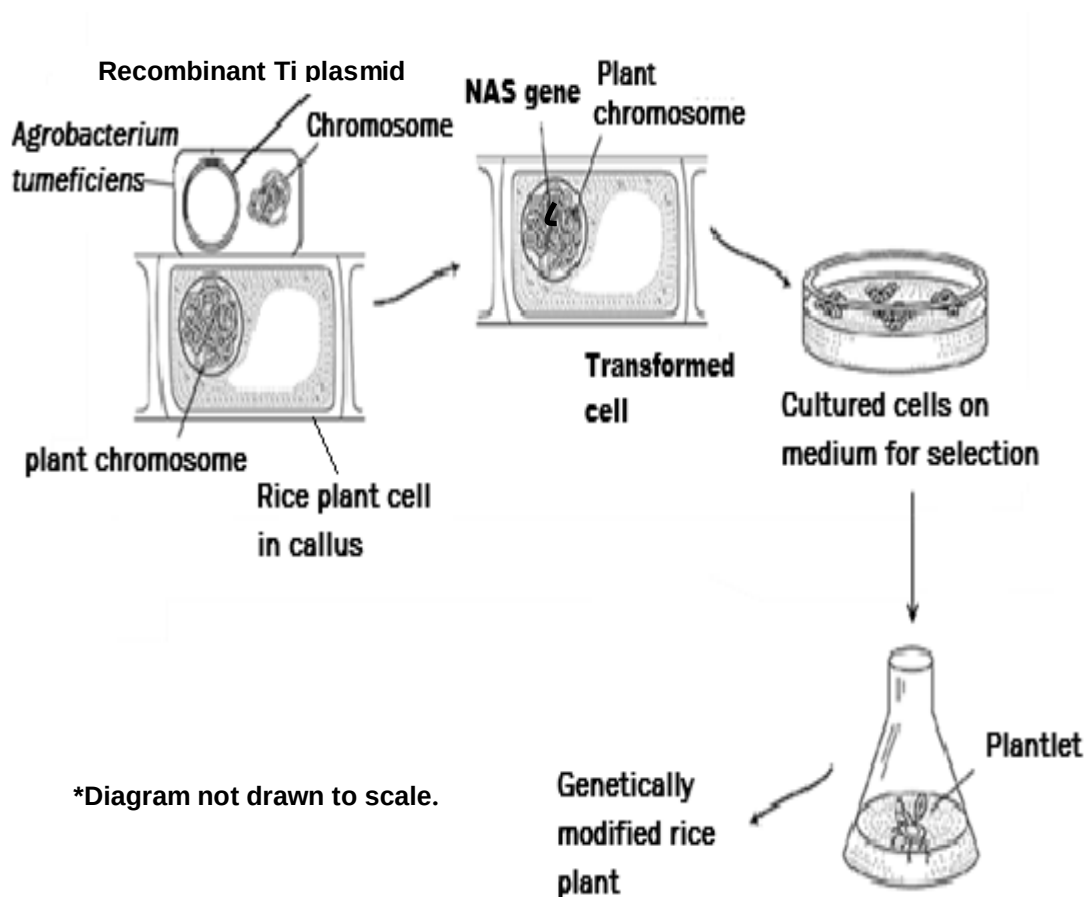


Figure 3.1

The naturally occurring Ti plasmid derived from *Agrobacterium tumefaciens*, is commonly used as a vector for introducing new foreign genes into plant cells. This plasmid integrates a segment of its DNA known as T-DNA, into the chromosomal DNA of host plant cells after infection. Figure 3.2 shows the structure of a natural Ti plasmid.

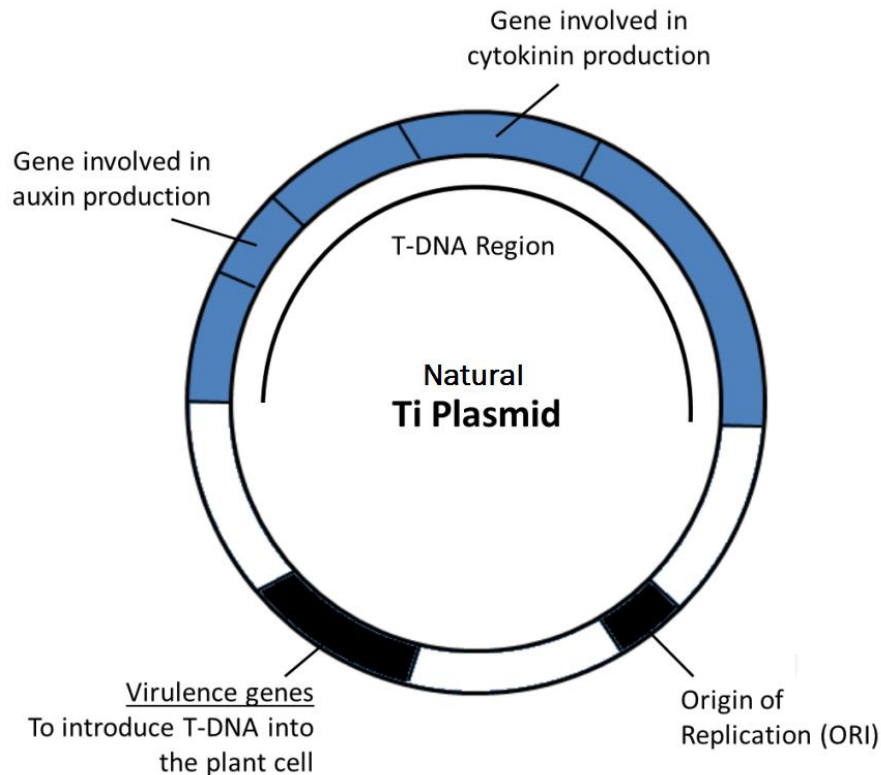


Figure 3.2

The genes for auxin and cytokinin are removed from the T-DNA region from the Ti plasmid before it can be used in the production of transgenic rice.

- (a) Explain why the genes for auxin and cytokinin are removed from the T-DNA region of the Ti Plasmid.

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Transgenic rice plants have to undergo a process of acclimatization before they can be transferred to a natural environment for growth.

(b)

(i) Explain the importance of acclimatization.

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(ii) Describe a key step in this process.

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A study was carried out to evaluate the efficacy of such transgenic rice with increased iron content. Twenty iron-deficient 12-year old girls were selected for the study. The haemoglobin count for each girl chosen was similarly below 12.8 gdl^{-3} . The normal haemoglobin count for girls at the age of 12 is 12.8 to 16.0 gdl^{-3} .

The girls selected for the study were randomly and equally sorted into a treatment group and a control group. The girls in the treatment group were requested to adhere to a diet of transgenic rice with increased iron content, three times a day, for 5 months. The girls in the control group were requested to adhere to a diet of normal rice grown in the same rice field, three times a day, for 5 months. The mean haemoglobin count for each group was tracked over this period.

The results of the analysis are shown in Figure 3.3.

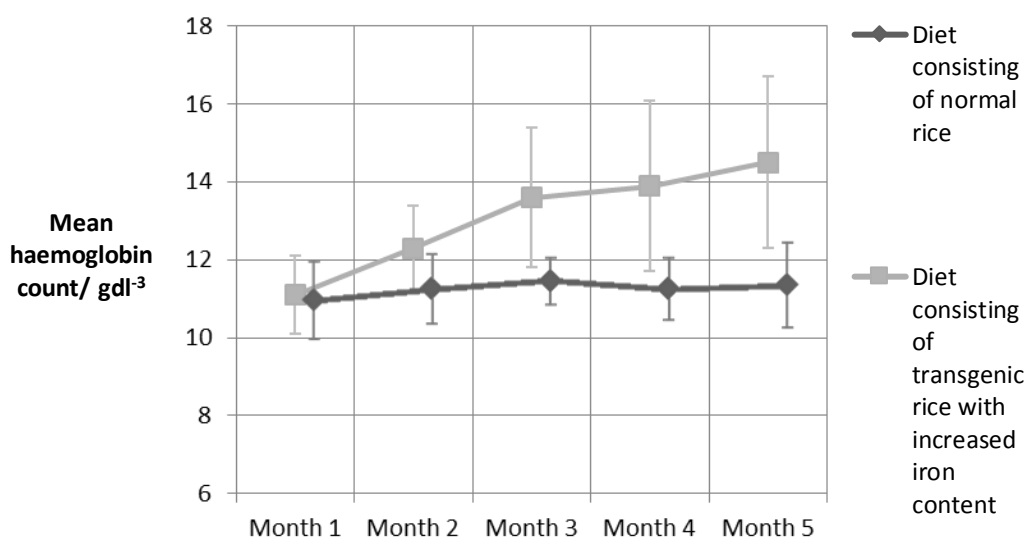


Figure 3.3

At the end of the study, the scientists involved concluded that the transgenic rice was more effective than normal rice in helping iron-deficient children.

(c) With reference to Figure 3.3, discuss the validity of this conclusion.

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In addition to producing more nutritious food, genetically engineered crops like Bt corn can also be used in solving the world demand for food.

(d) Explain how Bt corn helps solve the global food challenge.

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[Total: 11]

4. Beetroot is a starchy edible root from the *Beta vulgaris* plant whose cells contain a water-soluble red pigment, betacyanin, in their vacuoles. Beet root pigment is used commercially as a food dye and different methods have been used to extract the pigment from intact beetroot cells. This pigment cannot pass through membranes unless the membranes are damaged.

A colorimeter can be used to measure the absorbance of light at 550 nm by betacyanin solution. The concentration of betacyanin is proportional to its absorbance. The use of a colorimeter to measure absorbance by betacyanin is depicted in Figure 4.1.

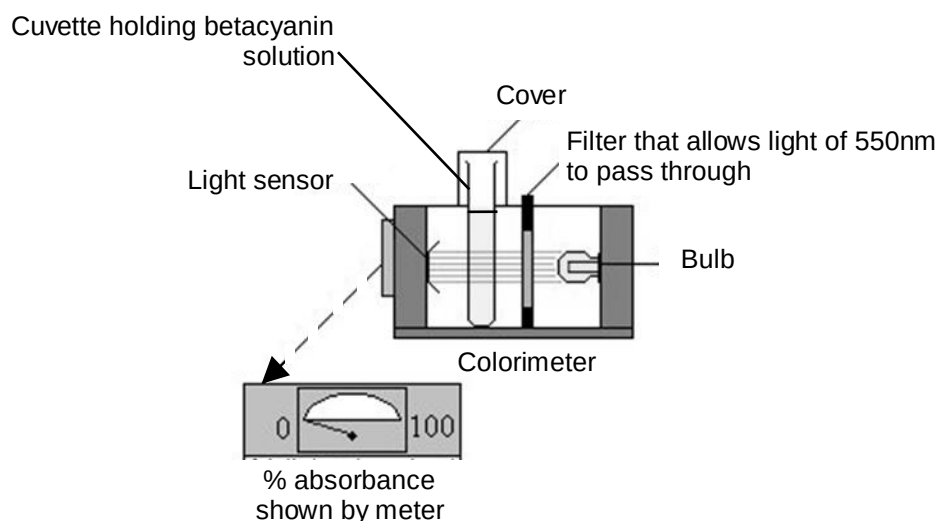


Fig. 4.1

Using the above information and your own knowledge, you are required to plan, but not carry out, an investigation into the effect of alcohol concentration on the permeability of the cell membrane of beetroot tissue.

Your planning must be based on the assumption that you have been provided with the following equipment and material which you must use.

- A large piece of beetroot tissue with the skin removed
- Sharp knife
- 10.0% Alcohol
- White tile
- Ruler
- Stopwatch
- Distilled water
- Colorimeter set at 550nm
- Cuvettes
- Normal laboratory glassware

Your plan should:

- have a clear and helpful structure such that the method you use can be repeated by anyone reading it,
- be illustrated by labelled diagrams, if necessary,
- identify independent and dependent variables,
- describe the method with scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results table(s) and graph(s) with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total: 12]

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Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5.

- (a)** Describe cystic fibrosis and explain how it can be treated using a non-viral gene delivery system. [8]
- (b)** Discuss the advantages and limitations of plant tissue culture. [6]
- (c)** Discuss the ethical concerns pertaining to the use of genetically modified animals. [6]

[Total: 20]

End Of Paper